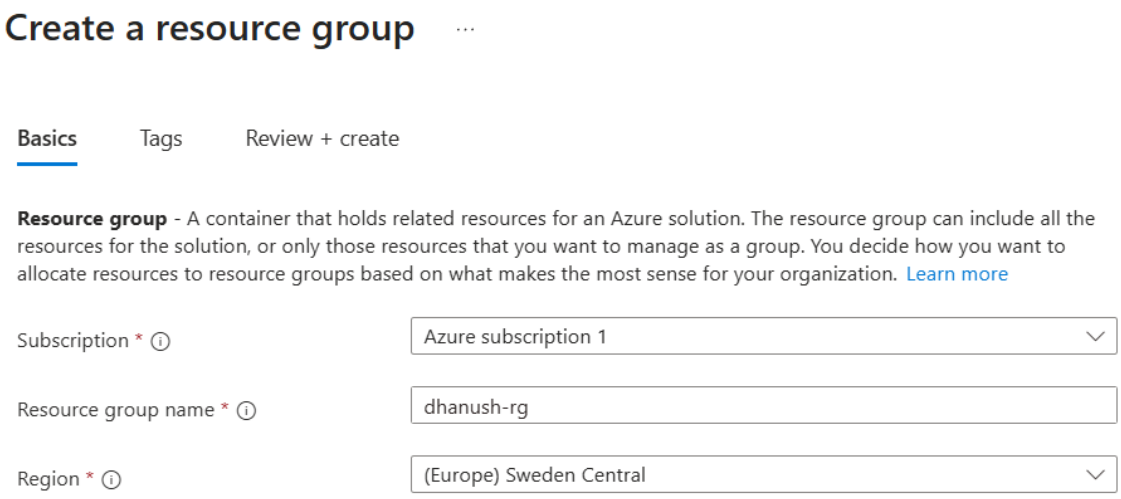
LAB 8 : Azure NSG Practice (Attach Later)

Objective

- Create 2 Linux VMs without NSG.
- Create NSG separately and attach it later.
- Test traffic before and after NSG.

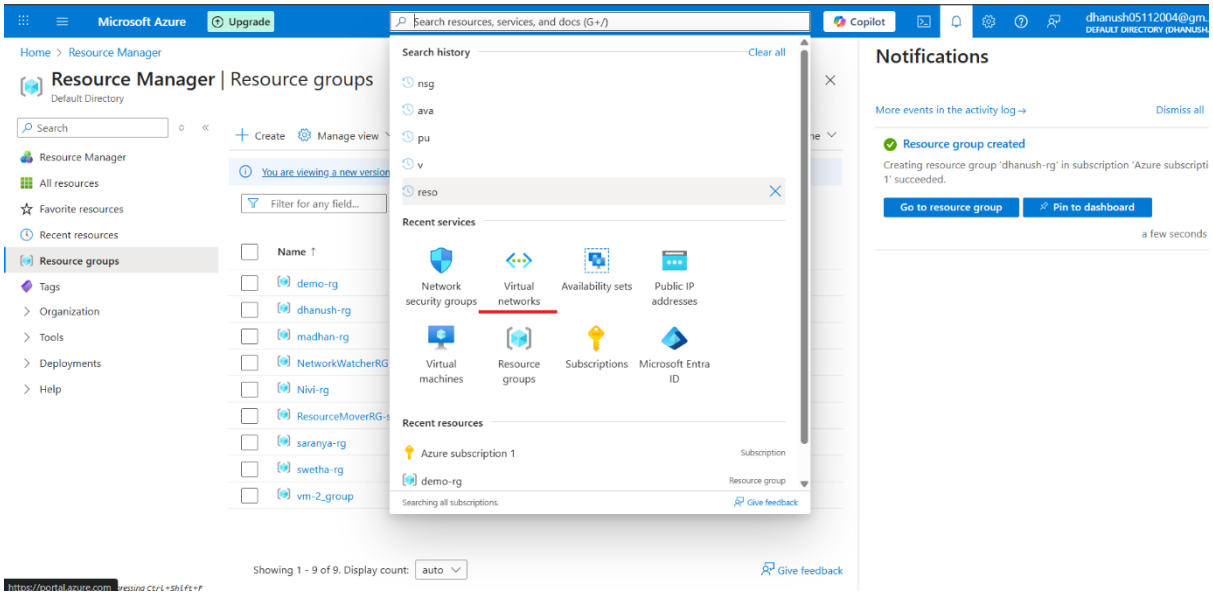
Step 1: Create Resource Group

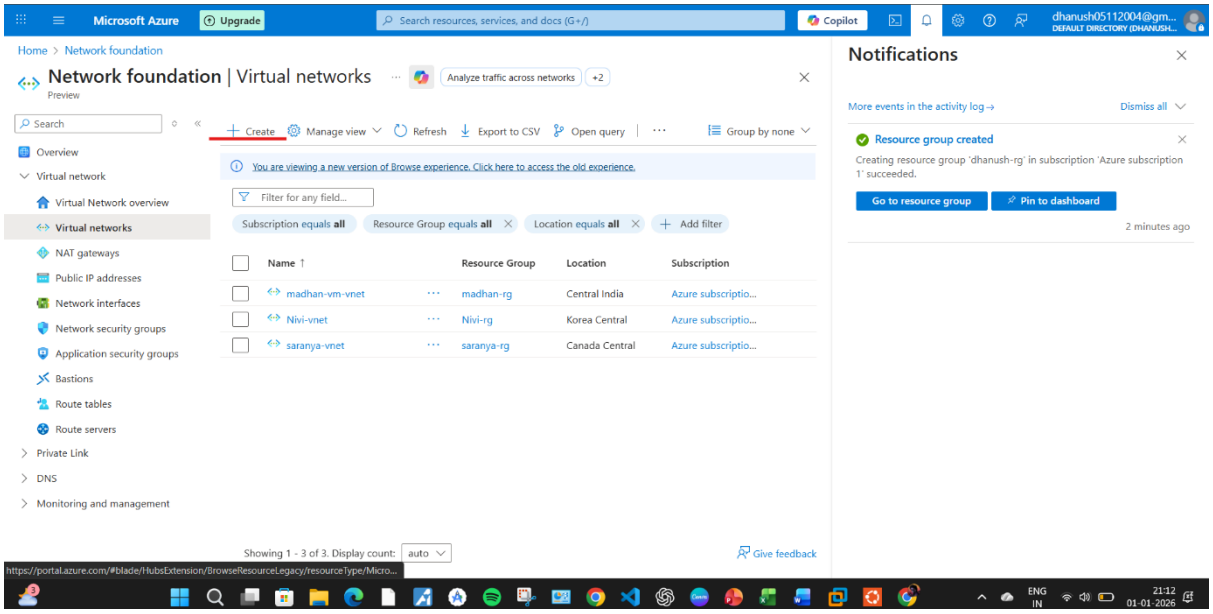
- Name: **studentname-rg**
- Region: East US (or your choice).

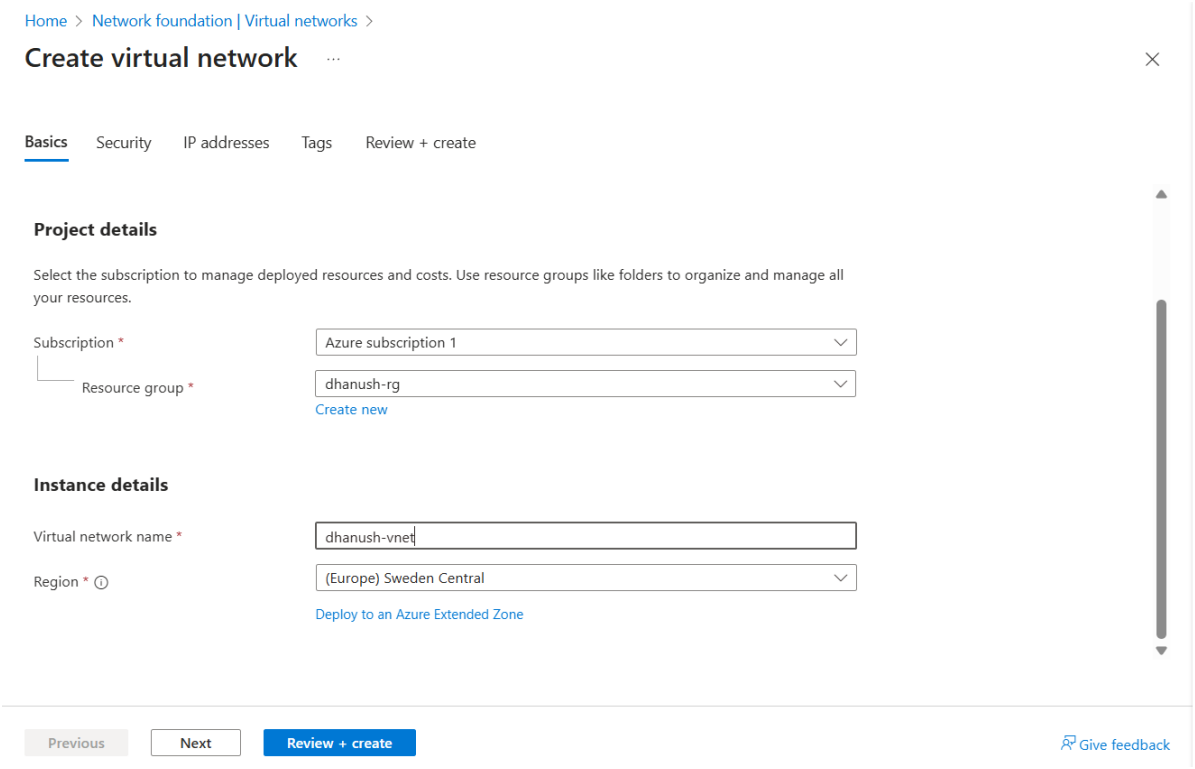


Step 2: Create Virtual Network

- Name: **studentname-vnet**
- Address space: 10.0.0.0/16

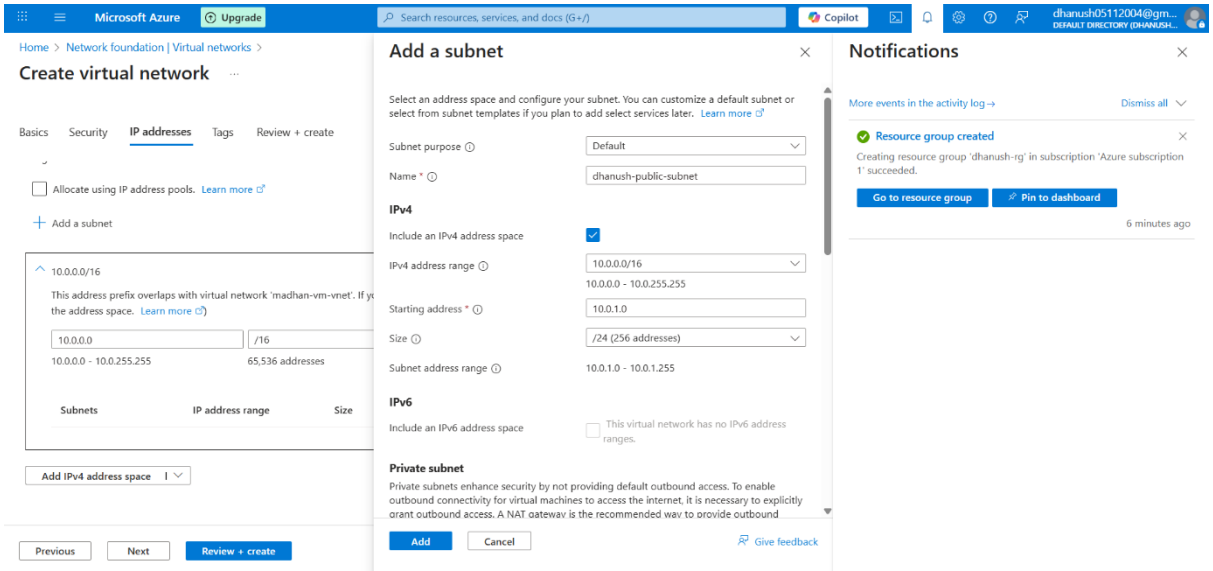

Microsoft Azure
Upgrade
Search resources, services, and docs (G+/)
Copilot
dhanush05112004@gm...
DEFAULT DIRECTORY (DHANUSH...
Home > Resource Manager
Resource Manager | Resource groups
Default Directory
Search
Create
Manage view
Resource Manager
All resources
Favorite resources
Recent resources
Resource groups
Tags
Organization
Tools
Deployments
Help
Filter for any field...
Name
demo-rg
dhanush-rg
madhan-rg
NetworkWatcherRG
Nivi-rg
saranya-rg
swetha-rg
vm-2_group
Showing 1 - 9 of 9. Display count: auto
Give feedback
Search history
Clear all
nsg
ava
pu
v
reso
Recent services
Network security groups
Virtual networks
Availability sets
Public IP addresses
Virtual machines
Resource groups
Subscriptions
Microsoft Entra ID
Recent resources
Azure subscription 1
Subscription
demo-rg
Resource group
Searching all subscriptions.
Give feedback
Notifications
More events in the activity log →
Dismiss all
Resource group created
Creating resource group 'dhanush-rg' in subscription 'Azure subscription 1' succeeded.
Go to resource group
Pin to dashboard
a few seconds


Microsoft Azure
Upgrade
Search resources, services, and docs (G+/)
Copilot
dhanush05112004@gm...
DEFAULT DIRECTORY (DHANUSH...
Home > Network foundation
Network foundation | Virtual networks
Preview
Analyze traffic across networks
+2
Search
Create
Manage view
Refresh
Export to CSV
Open query
Group by none
Overview
Virtual network
Virtual Network overview
Virtual networks
NAT gateways
Public IP addresses
Network interfaces
Network security groups
Application security groups
Bastions
Route tables
Route servers
Private Link
DNS
Monitoring and management
You are viewing a new version of Browse experience. Click here to access the old experience.
Filter for any field...
Subscription equals all
Resource Group equals all
Location equals all
Add filter
Name
Resource Group
Location
Subscription
madhan-vm-vnet
madhan-rg
Central India
Azure subscriptio...
Nivi-vnet
Nivi-rg
Korea Central
Azure subscriptio...
saranya-vnet
saranya-rg
Canada Central
Azure subscriptio...
Showing 1 - 3 of 3. Display count: auto
Give feedback
Notifications
More events in the activity log →
Dismiss all
Resource group created
Creating resource group 'dhanush-rg' in subscription 'Azure subscription 1' succeeded.
Go to resource group
Pin to dashboard
2 minutes ago



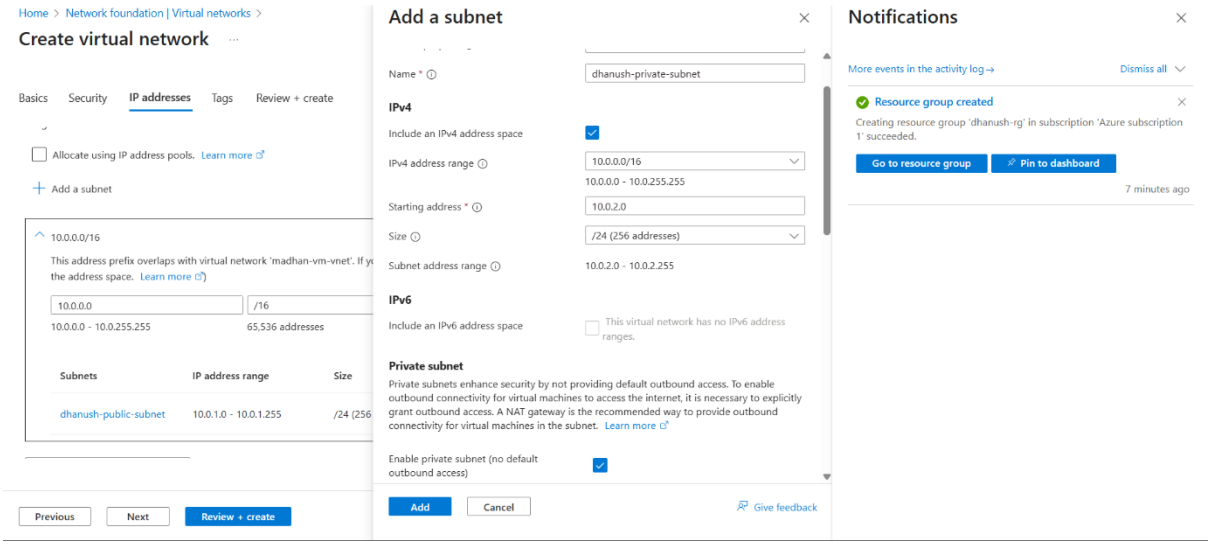
Subnet 1: FOR PUBLIC

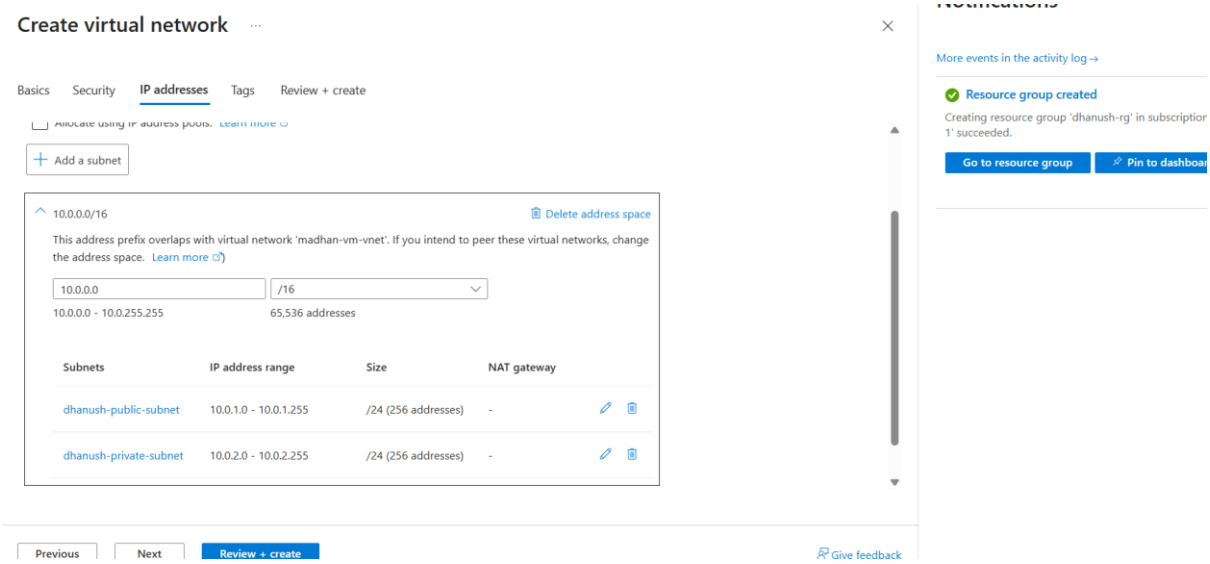
- Name: **studentname-subnet**
- Address range: 10.0.1.0/24

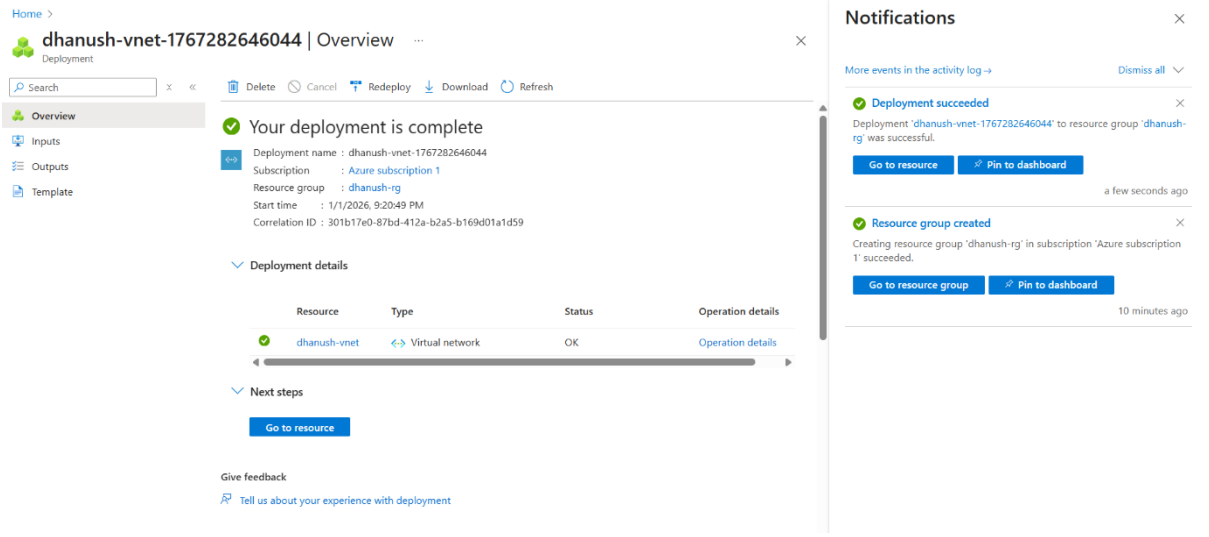


Subnet 2: FOR PRIVATE

- Name: **studentname-subnet**
- Address range: 10.0.2.0/24





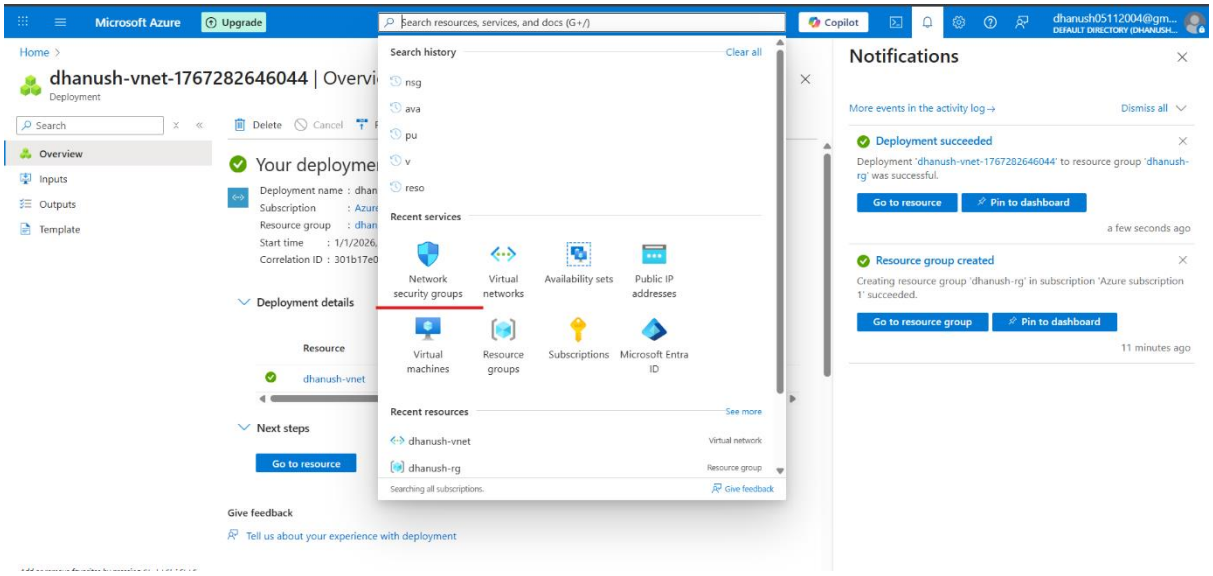


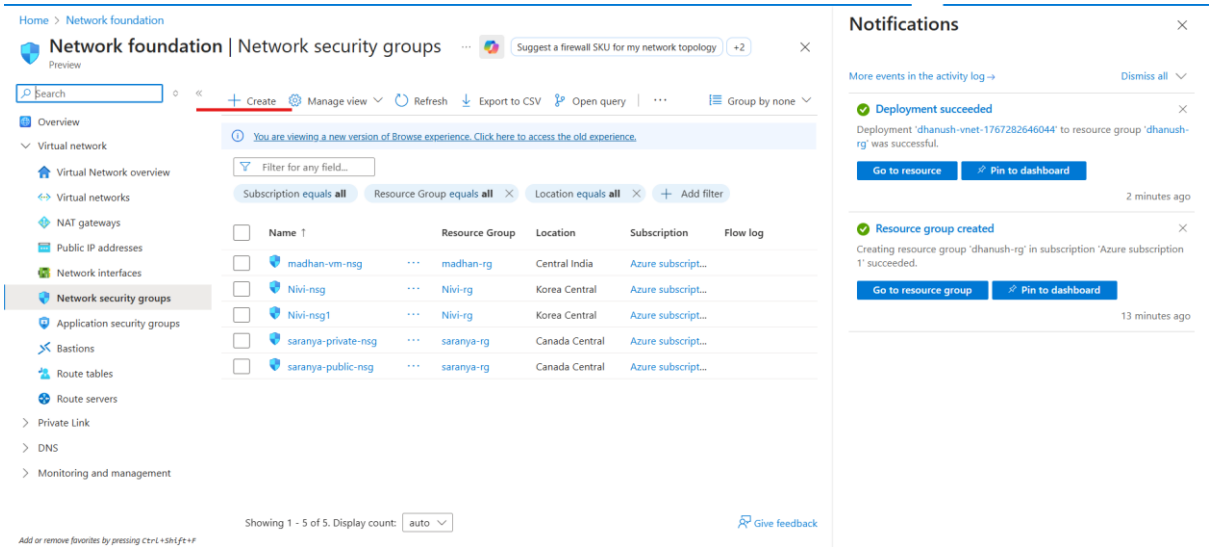
Step 3: Create NSG

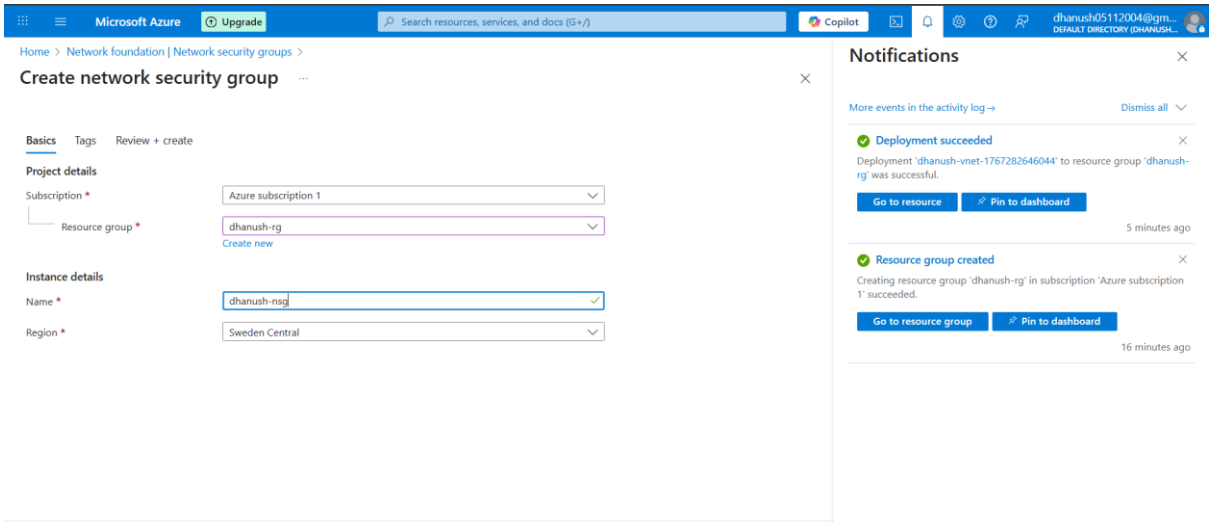
- **Name: studentname-nsg**

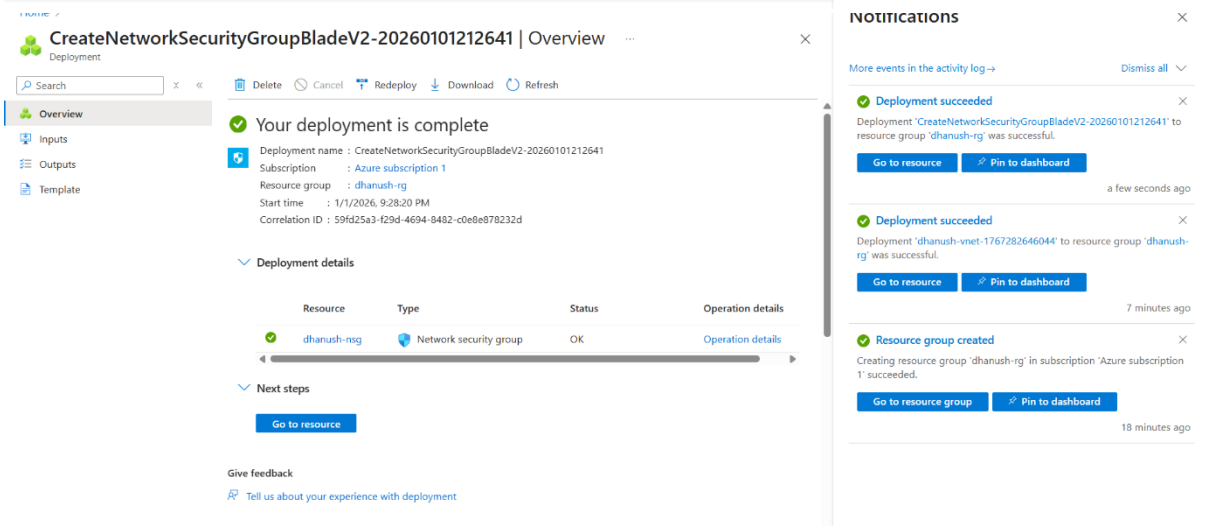
Add rules:

Rule Name	Priority	Protocol	Port	Action	Notes
studentname-allow-ssh	100	TCP	22	Allow	SSH
studentname-allow-ping	200	ICMP	Any	Allow	ICMP
studentname-deny-http	300	TCP	80	Deny	Block HTTP

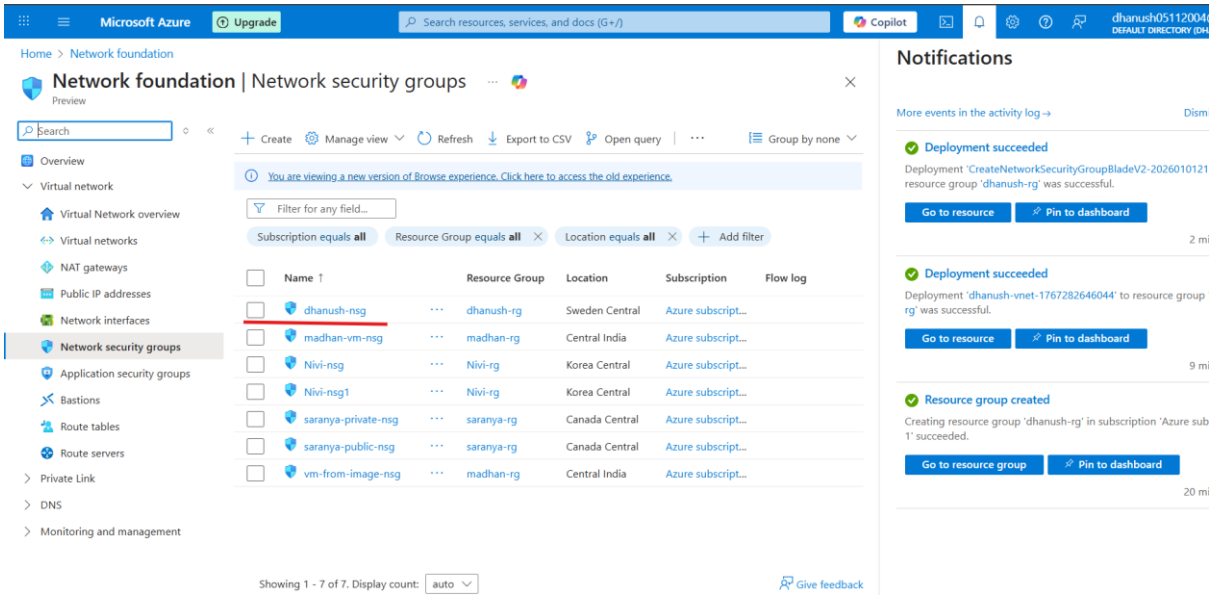

Microsoft Azure
Upgrade
Search resources, services, and docs (G+/)
Copilot
dhanush05112004@gm...
DEFAULT DIRECTORY (DHANUSH...
Home
dhanush-vnet-1767282646044 | Overvi
Deployment
Search
Overview
Inputs
Outputs
Template
Delete
Cancel
Your deployme
Deployment name : dhan
Subscription : Azure
Resource group : dhan
Start time : 1/1/2026,
Correlation ID : 301b17e0
Deployment details
Resource
dhanush-vnet
Next steps
Go to resource
Give feedback
Tell us about your experience with deployment
Search history
Clear all
nsg
ava
pu
v
reso
Recent services
Network security groups
Virtual networks
Availability sets
Public IP addresses
Virtual machines
Resource groups
Subscriptions
Microsoft Entra ID
Recent resources
See more
dhanush-vnet
Virtual network
dhanush-rg
Resource group
Searching all subscriptions.
Give feedback
Notifications
More events in the activity log →
Dismiss all
Deployment succeeded
Deployment 'dhanush-vnet-1767282646044' to resource group 'dhanush-rg' was successful.
Go to resource
Pin to dashboard
a few seconds ago
Resource group created
Creating resource group 'dhanush-rg' in subscription 'Azure subscription 1' succeeded.
Go to resource group
Pin to dashboard
11 minutes ago



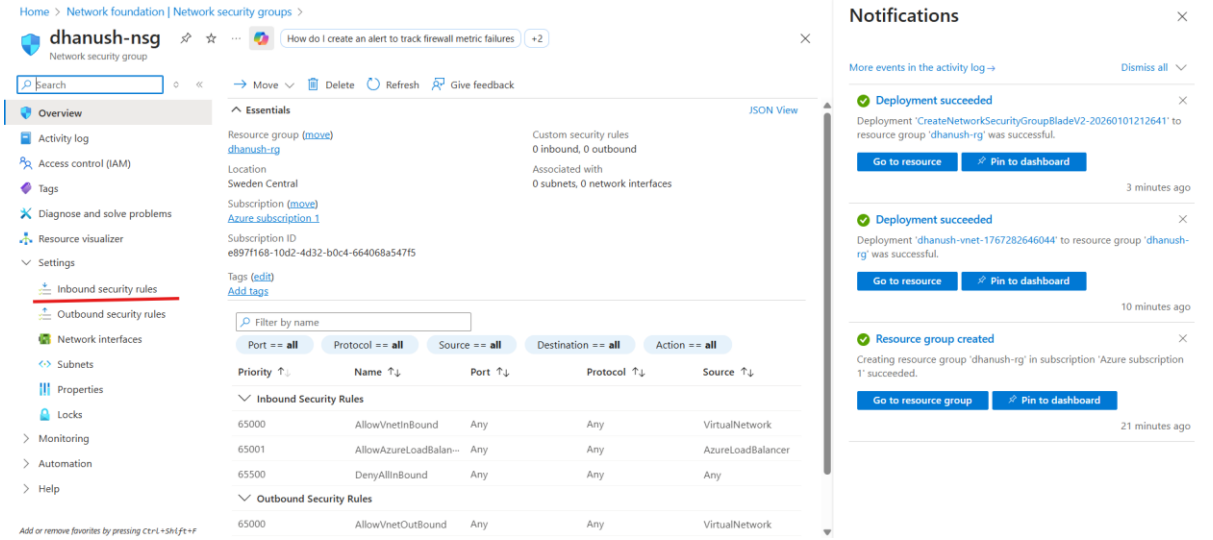


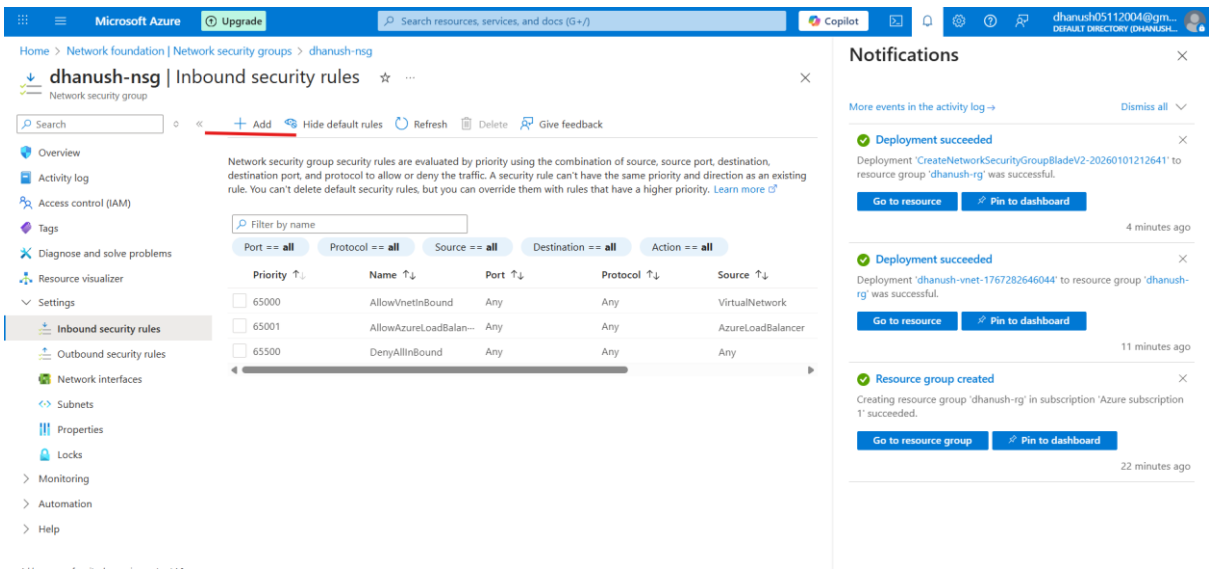


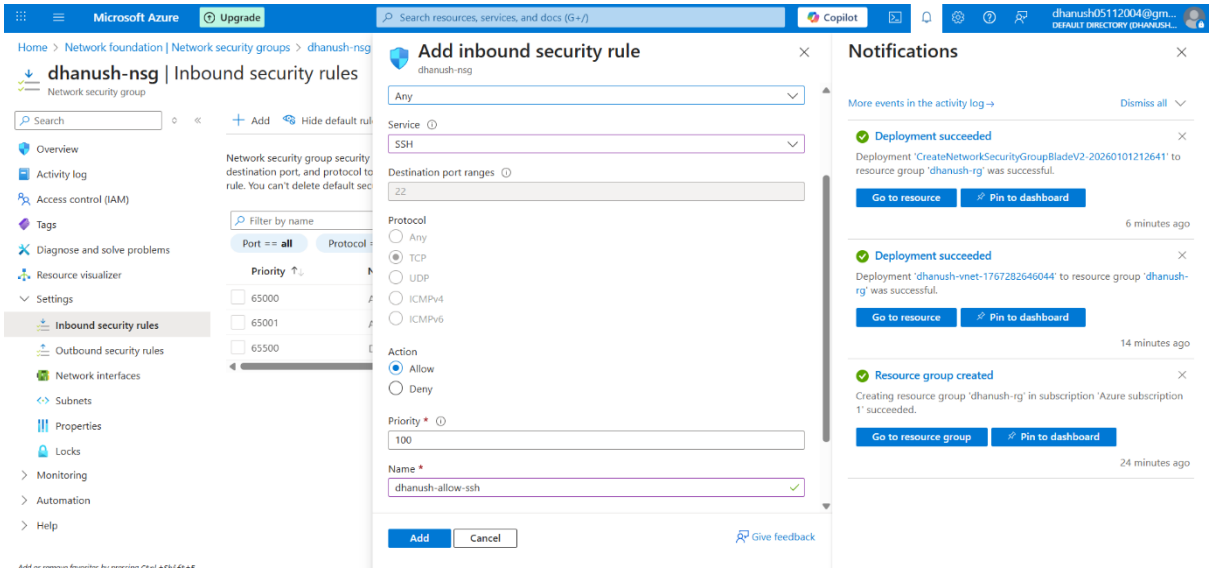
GO TO THE CREATED NSG

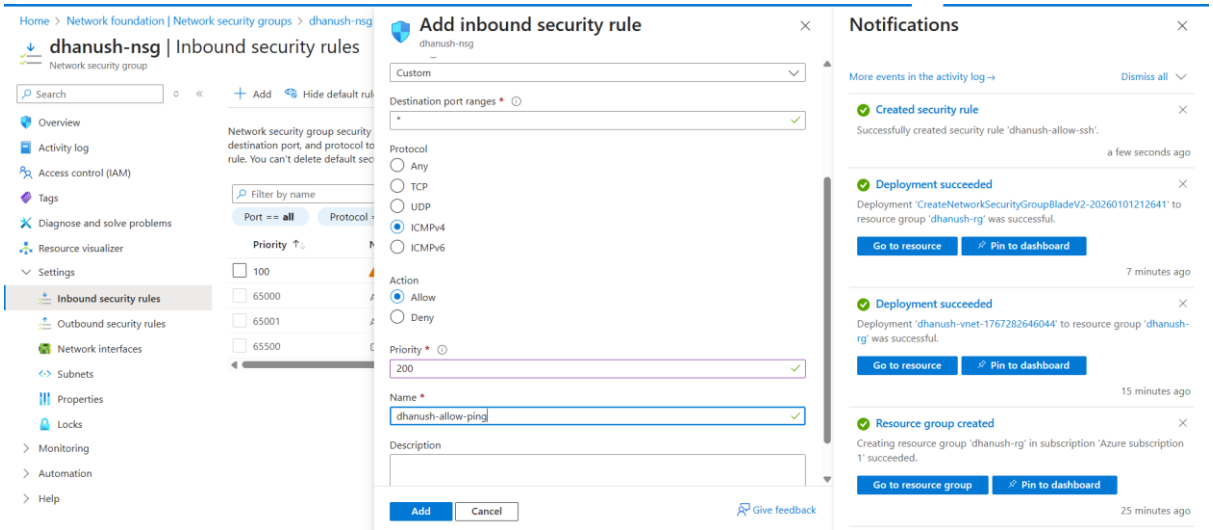


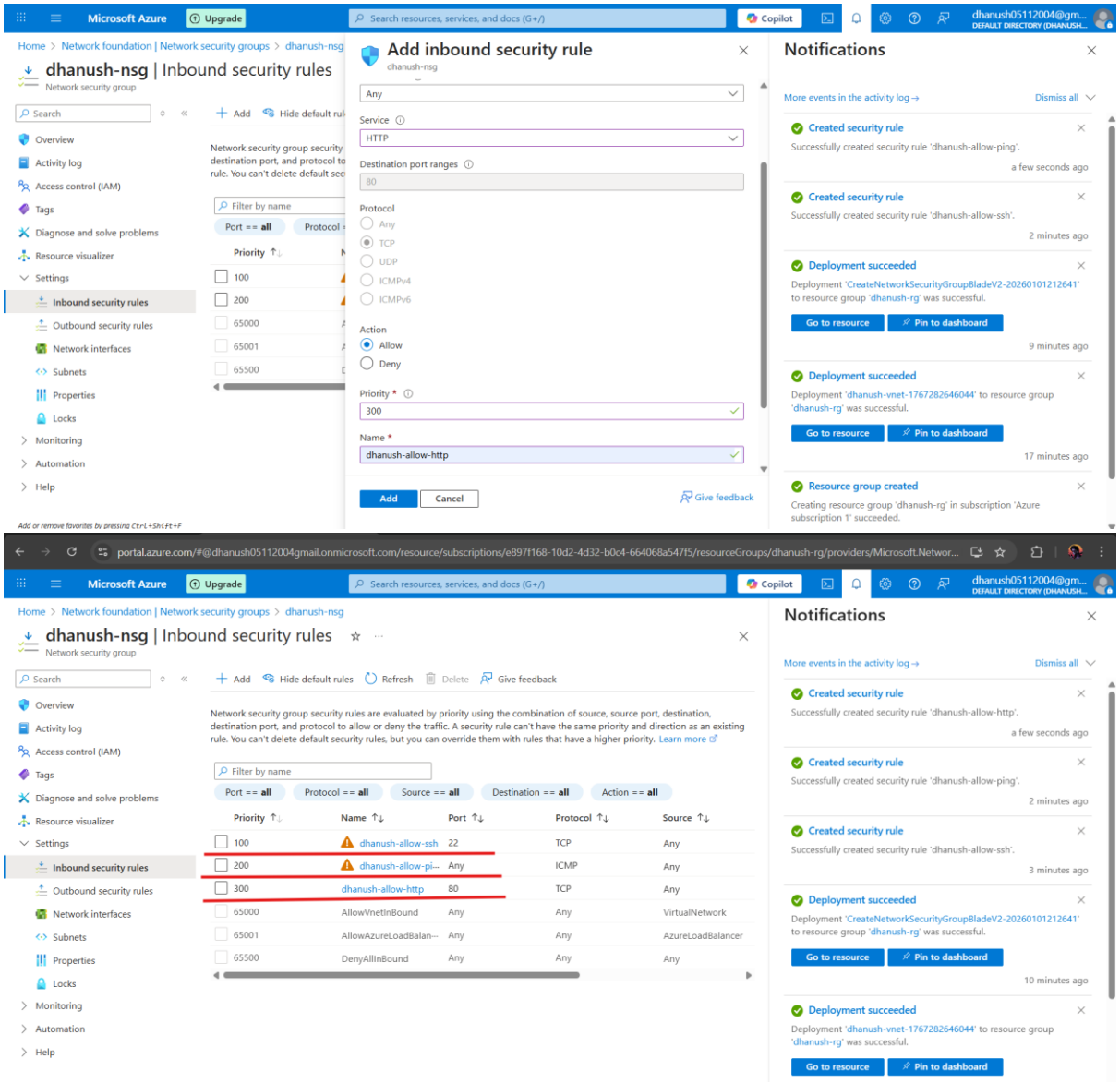
GO TO THE INBOUND SECURITY RULES






Microsoft Azure
Upgrade
Search resources, services, and docs (G+/)
Copilot
dhanush05112004@gm...
DEFAULT DIRECTORY (DHANUSH...
Home > Network foundation | Network security groups > dhanush-nsg
dhanush-nsg | Inbound security rules
Network security group
Search
Overview
Activity log
Access control (IAM)
Tags
Diagnose and solve problems
Resource visualizer
Settings
Inbound security rules
Outbound security rules
Network interfaces
Subnets
Properties
Locks
Monitoring
Automation
Help
Add
Hide default rul
Network security group security
destination port, and protocol to
rule. You can't delete default sec
Filter by name
Port == all
Protocol
Priority
65000
65001
65500
Add inbound security rule
dhanush-nsg
Any
Service
SSH
Destination port ranges
22
Protocol
Any
TCP
UDP
ICMPv4
ICMPv6
Action
Allow
Deny
Priority *
100
Name *
dhanush-allow-ssh
Add
Cancel
Give feedback
Notifications
More events in the activity log →
Dismiss all
Deployment succeeded
Deployment 'CreateNetworkSecurityGroupBladeV2-20260101212641' to resource group 'dhanush-rg' was successful.
Go to resource
Pin to dashboard
6 minutes ago
Deployment succeeded
Deployment 'dhanush-vnet-1767282646044' to resource group 'dhanush-rg' was successful.
Go to resource
Pin to dashboard
14 minutes ago
Resource group created
Creating resource group 'dhanush-rg' in subscription 'Azure subscription 1' succeeded.
Go to resource group
Pin to dashboard
24 minutes ago

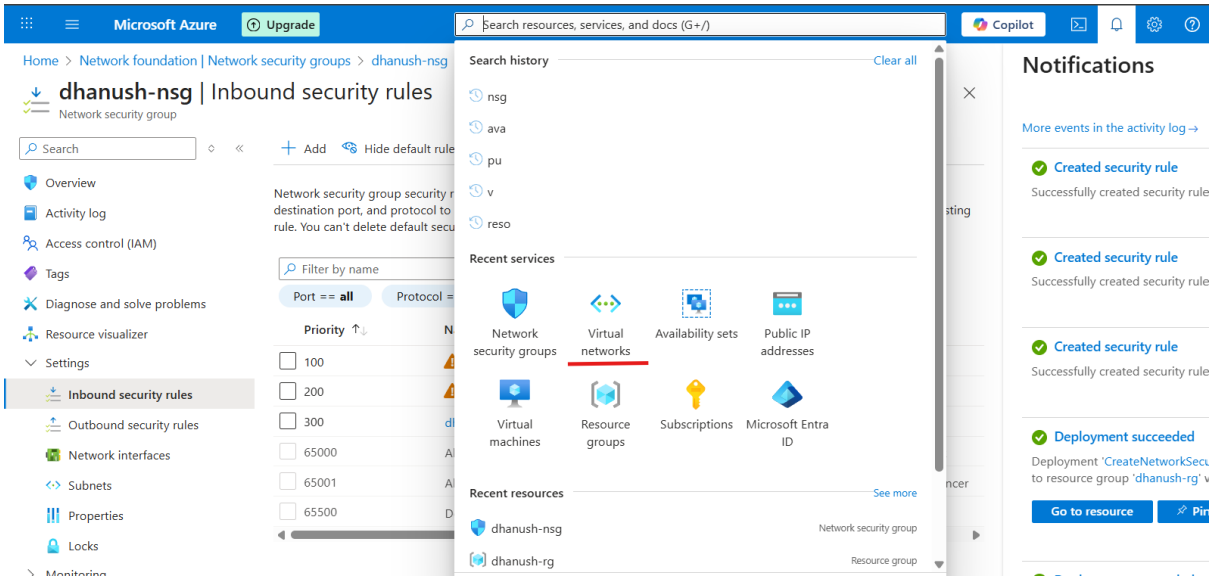

Home > Network foundation | Network security groups > dhanush-nsg
dhanush-nsg | Inbound security rules
Network security group
Search
Overview
Activity log
Access control (IAM)
Tags
Diagnose and solve problems
Resource visualizer
Settings
Inbound security rules
Outbound security rules
Network interfaces
Subnets
Properties
Locks
Monitoring
Automation
Help
Add
Hide default rul
Network security group security
destination port, and protocol to
rule. You can't delete default sec
Filter by name
Port == all
Protocol
Priority
100
65000
65001
65500
Add inbound security rule
dhanush-nsg
Custom
Destination port ranges *
*
Protocol
Any
TCP
UDP
ICMPv4
ICMPv6
Action
Allow
Deny
Priority *
200
Name *
dhanush-allow-ping
Description
Add
Cancel
Give feedback
Notifications
More events in the activity log →
Dismiss all
Created security rule
Successfully created security rule 'dhanush-allow-ssh'.
a few seconds ago
Deployment succeeded
Deployment 'CreateNetworkSecurityGroupBladeV2-20260101212641' to resource group 'dhanush-rg' was successful.
Go to resource
Pin to dashboard
7 minutes ago
Deployment succeeded
Deployment 'dhanush-vnet-1767282646044' to resource group 'dhanush-rg' was successful.
Go to resource
Pin to dashboard
15 minutes ago
Resource group created
Creating resource group 'dhanush-rg' in subscription 'Azure subscription 1' succeeded.
Go to resource group
Pin to dashboard
25 minutes ago

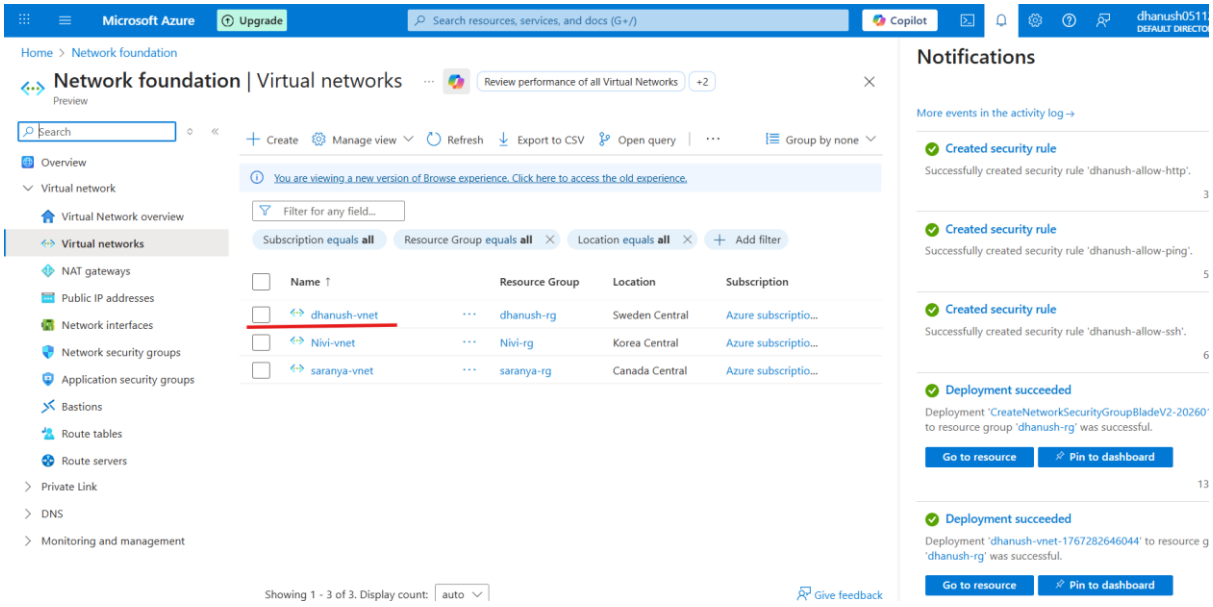


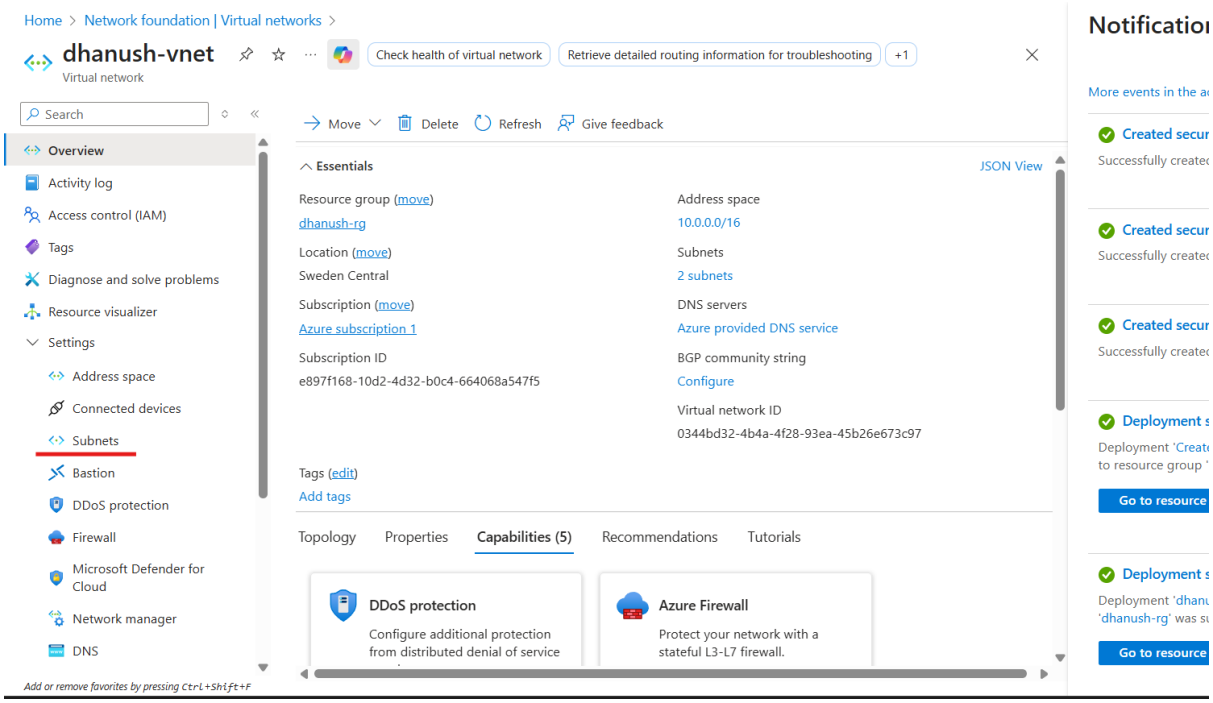
Step 4: Associate NSG

Go to the vnet and click subnet

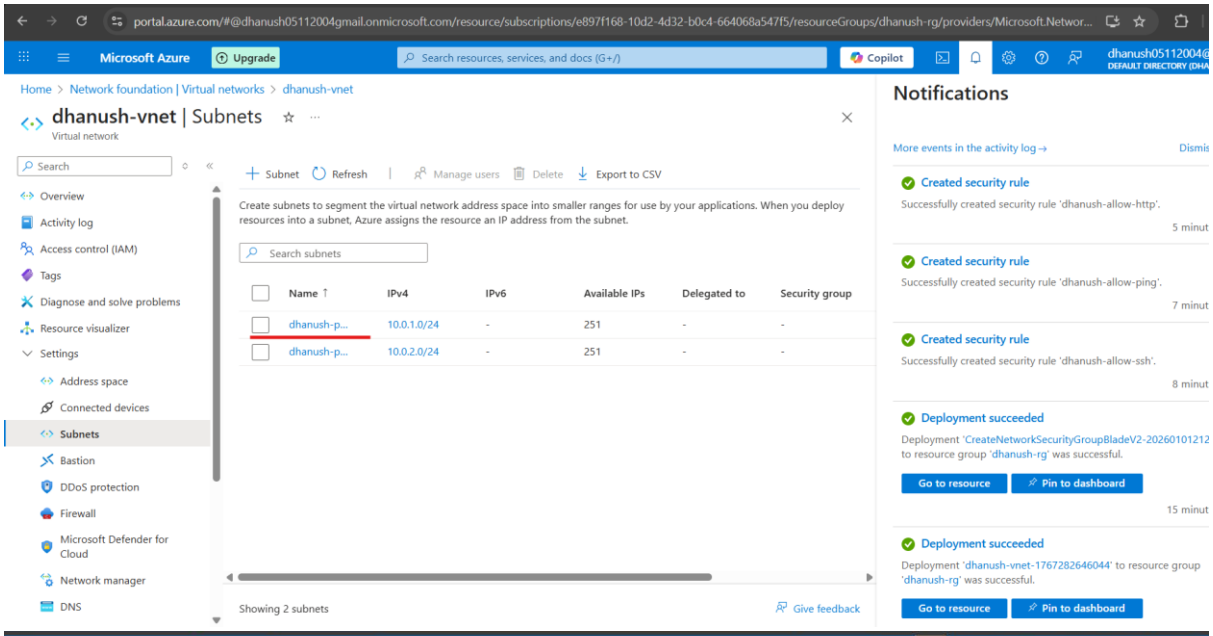
- Go to **studentname-subnet** → Associate NSG → Select **studentname-nsg**.

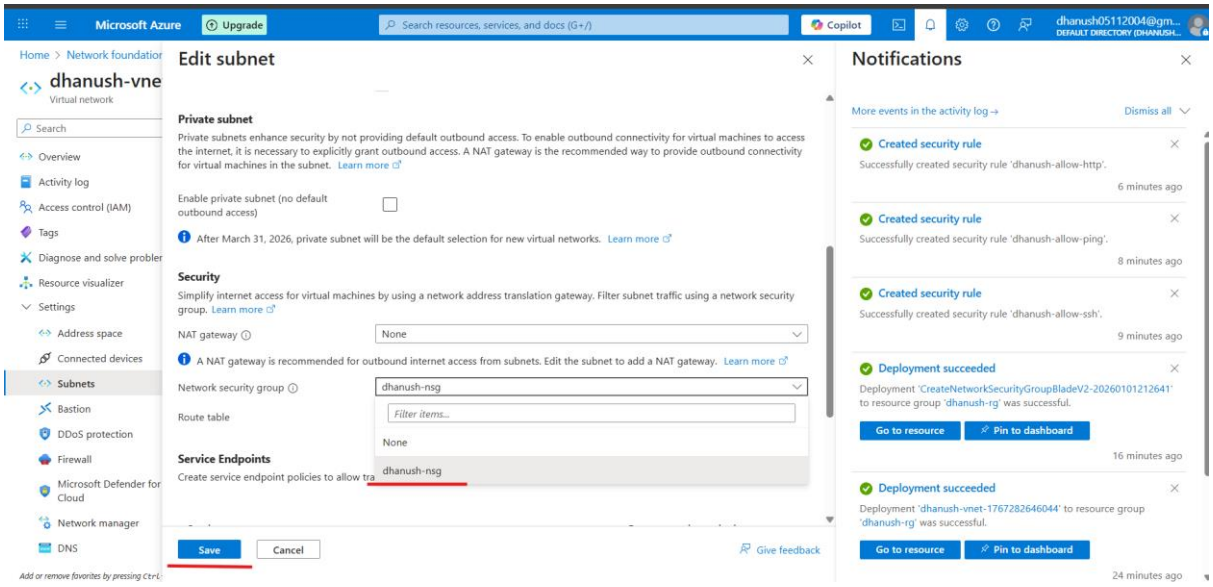

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Upgrade
Search resources, services, and docs (G+/)
Copilot
Home > Network foundation | Network security groups > dhanush-nsg
dhanush-nsg | Inbound security rules
Network security group
Search
Add
Hide default rule
Overview
Activity log
Access control (IAM)
Tags
Diagnose and solve problems
Resource visualizer
Settings
Inbound security rules
Outbound security rules
Network interfaces
Subnets
Properties
Locks
Filter by name
Port == all
Protocol =
Priority
100
200
300
65000
65001
65500
Search history
Clear all
nsg
ava
pu
v
reso
Recent services
Network security groups
Virtual networks
Availability sets
Public IP addresses
Virtual machines
Resource groups
Subscriptions
Microsoft Entra ID
Recent resources
See more
dhanush-nsg
Network security group
dhanush-rg
Resource group
Notifications
More events in the activity log →
Created security rule
Successfully created security rule
Created security rule
Successfully created security rule
Created security rule
Successfully created security rule
Deployment succeeded
Go to resource


Microsoft Azure
Upgrade
Search resources, services, and docs (G+/)
Copilot
Home > Network foundation
Network foundation | Virtual networks
Preview
Review performance of all Virtual Networks
+2
Search
Overview
Virtual network
Virtual Network overview
Virtual networks
NAT gateways
Public IP addresses
Network interfaces
Network security groups
Application security groups
Bastions
Route tables
Route servers
Private Link
DNS
Monitoring and management
Create
Manage view
Refresh
Export to CSV
Open query
Group by none
You are viewing a new version of Browse experience. Click here to access the old experience.
Filter for any field...
Subscription equals all
Resource Group equals all
Location equals all
Add filter
Name
Resource Group
Location
Subscription
dhanush-vnet
dhanush-rg
Sweden Central
Azure subscriptio...
Nivi-vnet
Nivi-rg
Korea Central
Azure subscriptio...
saranya-vnet
saranya-rg
Canada Central
Azure subscriptio...
Showing 1 - 3 of 3. Display count: auto
Give feedback
Notifications
More events in the activity log →
Created security rule
Successfully created security rule 'dhanush-allow-http'.
Created security rule
Successfully created security rule 'dhanush-allow-ping'.
Created security rule
Successfully created security rule 'dhanush-allow-ssh'.
Deployment succeeded
Deployment 'CreateNetworkSecurityGroupBladeV2-20260' to resource group 'dhanush-rg' was successful.
Go to resource
Pin to dashboard
Deployment succeeded
Deployment 'dhanush-vnet-1767282646044' to resource g 'dhanush-rg' was successful.
Go to resource
Pin to dashboard

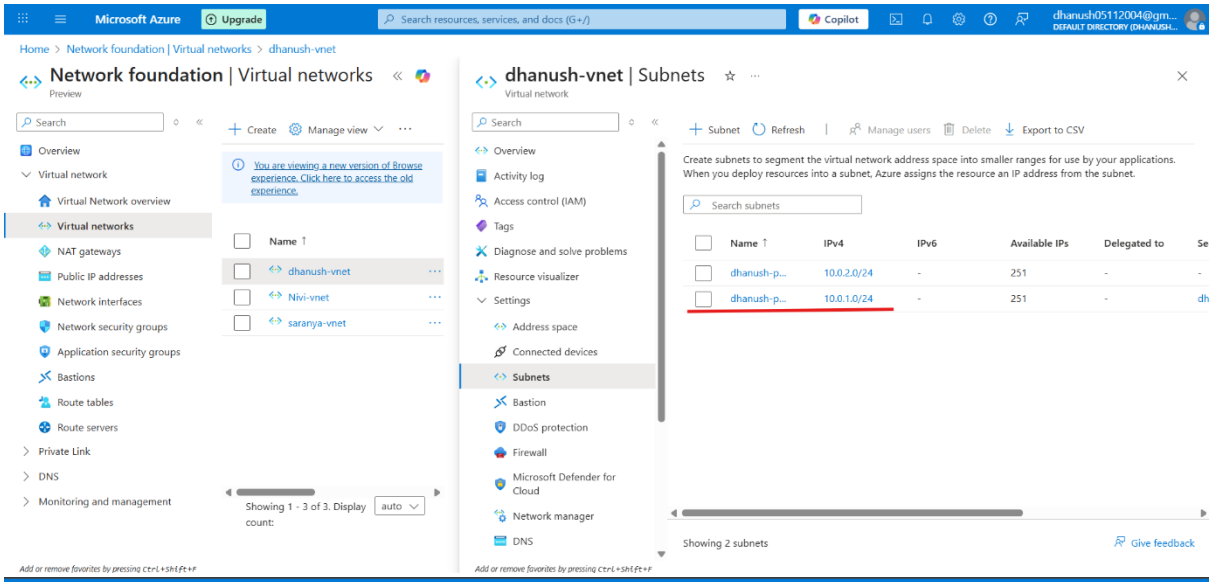


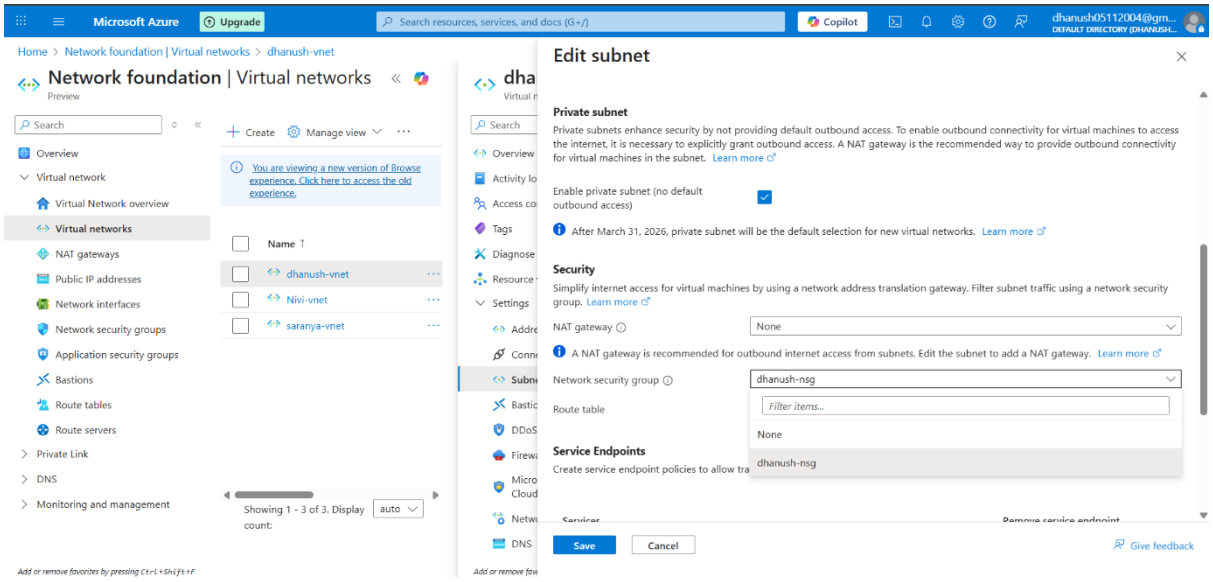
Associate with public subnet





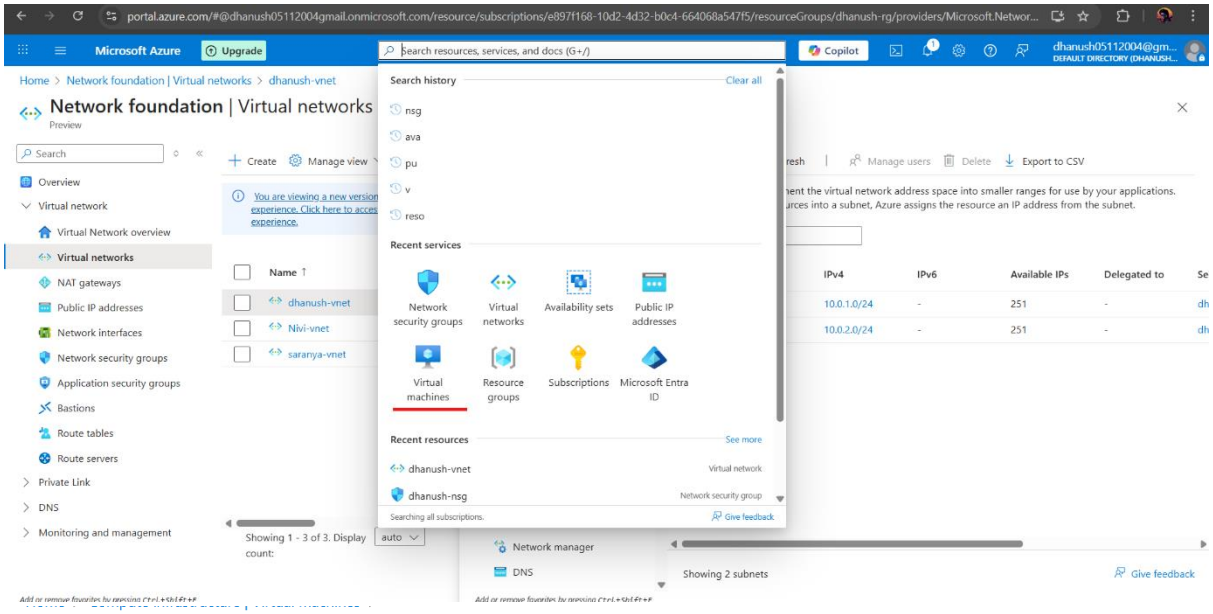
Associate with private subnet





Step 5: Create 2 VMs

VM1 (public access)



Create a virtual machine

Help me create a VM optimized for high availability | Help me create a low cost VM | Help me choose the right VM size for my workload

Help me create a low cost VM | Help me create a VM optimized for high availability | Help me choose the right VM size for my workload

This subscription may not be eligible to deploy VMs of certain sizes in certain regions.

Project details

Select the subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all your resources.

Subscription * : Azure subscription 1

Resource group * : dhanush-rg

Create new

Instance details

Virtual machine name * : dhanush-vm1

Region * : (Europe) Sweden Central

Deploy to an Azure Extended Zone

Availability options : Availability zone

Zone options :

- (●) Self-selected zone
  Choose up to 3 availability zones, one VM per zone
- ( ) Azure-selected zone (Preview)

< Previous | Next : Disks > | Review + create

You are in the free trial period. Costs associated with this VM can be covered by any remaining credits on your subscription. Learn more

Size * : Standard_D2_v4 - 2 vcpus, 8 GiB memory ($74.46/month)
See all sizes

Enable Hibernation : Hibernate does not currently support Trusted launch and Confidential virtual machines for Linux images. Learn more

Administrator account

Authentication type : SSH public key / Password

Username * : dhanushuser

Password * : ••••••••••••

Confirm password * : ••••••••••••

Inbound port rules

Select which virtual machine network ports are accessible from the public internet. You can specify more limited or granular

< Previous | Next : Disks > | Review + create

Inbound port rules

Select which virtual machine network ports are accessible from the public internet. You can specify more limited or granular network access on the Networking tab.

Public inbound ports * : None / Allow selected ports

Select inbound ports : Select one or more ports

All traffic from the internet will be blocked by default. You will be able to change inbound port rules in the VM > Networking page.

Create a virtual machine

Help me create a VM optimized for high availability | Help me create a low cost VM | Help me choose the right VM

Help me create a low cost VM | Help me create a VM optimized for high availability | Help me choose the right VM size for my workload

Network interface

When creating a virtual machine, a network interface will be created for you.

Virtual network * : dhanush-vnet
Create new

Subnet * : dhanush-public-subnet (10.0.1.0/24)
Manage subnet configuration

Public IP : (new) dhanush-vm1-ip
Create new

Public IP addresses have a nominal charge. Estimate price

NIC network security group : None / Basic / Advanced

The selected subnet 'dhanush-public-subnet (10.0.1.0/24)' is already associated to a network security group 'dhanush-nsg'. We recommend managing connectivity to this virtual machine via the existing network security group instead of creating a new one here.

Configure network security group * : dhanush-nsg
Create new

< Previous | Next : Management > | Review + create

Create a virtual machine

Help me create a VM optimized for high availability | Help me create a low cost VM | Help me choose the right VM size for my

Validation passed

Help me create a low cost VM | Help me create a VM optimized for high availability | Help me choose the right VM size for my workload

Basics | Disks | Networking | Management | Monitoring | Advanced | Tags | **Review + create**

Price

1 X Standard D2 v4
by Microsoft
Terms of use | Privacy policy

Subscription credits apply

0.1020 USD/hr
Pricing for other VM sizes

TERMS

By clicking "Create", I (a) agree to the legal terms and privacy statement(s) associated with the Marketplace offering(s) listed above; (b) authorize Microsoft to bill my current payment method for the fees associated with the offering(s), with the same billing frequency as my Azure subscription; and (c) agree that Microsoft may share my contact, usage and transactional information with the provider(s) of the offering(s) for support, billing and other transactional activities. Microsoft does not provide rights for third-party offerings. See the Azure Marketplace Terms for additional details.

Name: dhanush v

Preferred e-mail address: dhanush05112004@gmail.com

< Previous | Next > | Create

The selected subnet 'dhanush-public-subnet (10.0.1.0/24)' is already associated to a network security group 'dhanush-nsg'. We recommend managing connectivity to this virtual machine via the existing network security group instead of creating a new one here.

Configure network security group *: dhanush-nsg
Create new

Delete public IP and NIC when VM is deleted

Enable accelerated networking

Load balancing

You can place this virtual machine in the backend pool of an existing Azure load balancing solution. Learn more

Load balancing options

- None
- Azure load balancer
  Supports all TCP/UDP network traffic, port-forwarding, and outbound flows.
- Application gateway
  Web traffic load balancer for HTTP/HTTPS with URL-based routing, SSL termination, session persistence, and web application firewall.

< Previous | Next : Management > | Review + create

Home >

CreateVm-almalinux.almalinux-x86_64-9-gen2-20260101215506 | Overview

Deployment

Search

Overview
Inputs
Outputs
Template

Delete | Cancel | Redeploy | Download | Refresh

Your deployment is complete

Deployment name: CreateVm-almalinux.almalinux-x86_64-9-gen2-...
Subscription: Azure subscription 1
Resource group: dhanush-rg

Start time: 1/1/2026, 10:01:48 PM
Correlation ID: 60a821fc-9409-423b-a17d-b6231e9c0f98

Deployment details

Resource	Type	Status	Operation details
dhanush-vm1	Microsoft.Compute/virtualMachin...	OK	Operation details
dhanush-vm1863_z1	Microsoft.Network/networkInterf...	OK	Operation details
dhanush-vm1-ip	Microsoft.Network/publicIpAddre...	OK	Operation details

Next steps

Setup auto-shutdown Recommended
Monitor VM health, performance and network dependencies Recommended
Run a script inside the virtual machine Recommended

Go to resource | Create another VM

Give feedback
Tell us about your experience with deployment

Add or remove favorites by pressing Ctrl+Shift+F

Deployment succeeded
Deployment 'CreateVm-almalinux.almalinux-x86_64-9-gen2-20260101215506' to resource group 'dhanush-rg' was successful.
Go to resource | Pin to dashboard

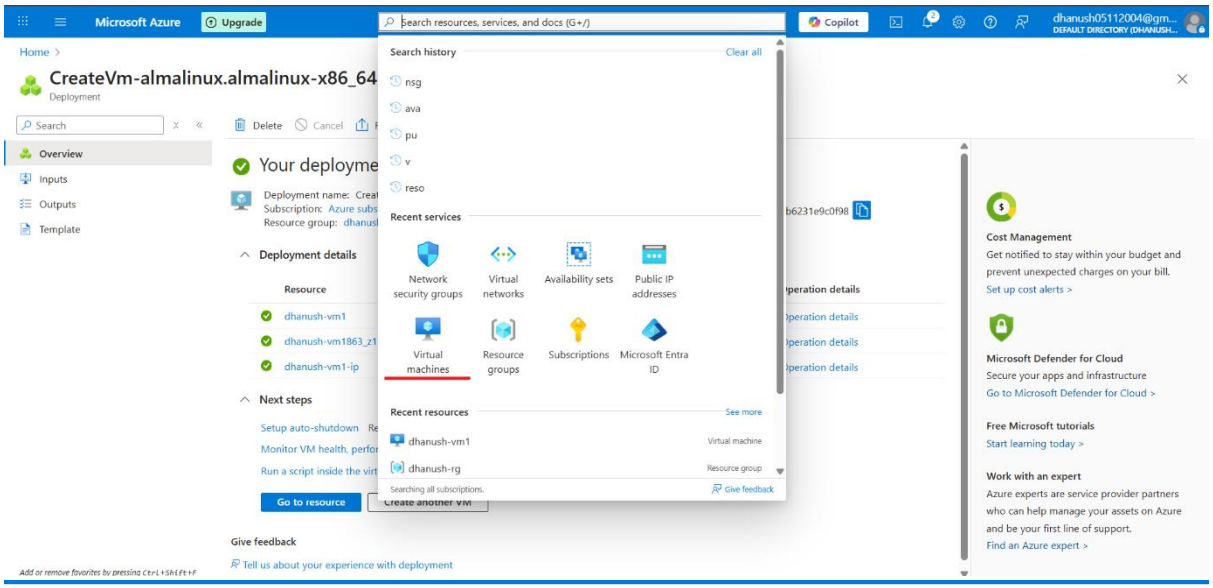
Cost Management
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VM2 (private only)



Home > Compute infrastructure | Virtual machines >

Create a virtual machine

Help me choose the right VM size for my workload | Help me create a low cost VM | Help me create a VM optimized for high availability

Help me create a low cost VM | Help me create a VM optimized for high availability | Help me choose the right VM size for my workload

Project details

Select the subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all your resources.

Subscription * : Azure subscription 1

Resource group * : dhanush-rg

Create new

Instance details

Virtual machine name * : dhanush-vm2

Region * : (Europe) Sweden Central

Deploy to an Azure Extended Zone

Availability options : Availability zone

Zone options :

- (●) Self-selected zone
  Choose up to 3 availability zones, one VM per zone
- ( ) Azure-selected zone (Preview)
  Let Azure assign the best zone for your needs

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Administrator account

Authentication type
- ○ SSH public key
- ◉ Password

Username * dhanushuser

Password * ••••••••••••

Confirm password * ••••••••••••

Inbound port rules

Select which virtual machine network ports are accessible from the public internet. You can specify more limited or granular network access on the Networking tab.

Public inbound ports *
- ◉ None
- ○ Allow selected ports

Select inbound ports: Select one or more ports

All traffic from the internet will be blocked by default. You will be able to change inbound port rules in the VM > Networking page.

< Previous | Next : Disks > | Review + create

Create a virtual machine

Help me create a low cost VM | Help me create a VM optimized for high availability | Help me choose the right VM size for my worklo

Help me create a low cost VM | Help me create a VM optimized for high availability | Help me choose the right VM size for my workload

Network interface

When creating a virtual machine, a network interface will be created for you.

Virtual network * dhanush-vnet

Create new

Subnet * dhanush-private-subnet (10.0.2.0/24)

Manage subnet configuration

Public IP: None

- (new) dhanush-vm2-ip
- None

NIC network security group
- Basic
- ○ Advanced

The selected subnet 'dhanush-private-subnet (10.0.2.0/24)' is already associated to a network security group 'dhanush-nsg'. We recommend managing connectivity to this virtual machine via the existing network security group instead of creating a new one here.

Delete NIC when VM is deleted ☐

Enable accelerated networking ☑

< Previous | Next : Management > | Review + create

Create a virtual machine

Help me create a low cost VM | Help me create a VM optimized for high availability | Help me choose the right VM size fo

Help me create a low cost VM | Help me create a VM optimized for high availability | Help me choose the right VM size for my workload

Subnet * ⓘ
dhanush-private-subnet (10.0.2.0/24)
Manage subnet configuration

Public IP ⓘ
None
Create new

NIC network security group ⓘ
- None
- Basic
- Advanced

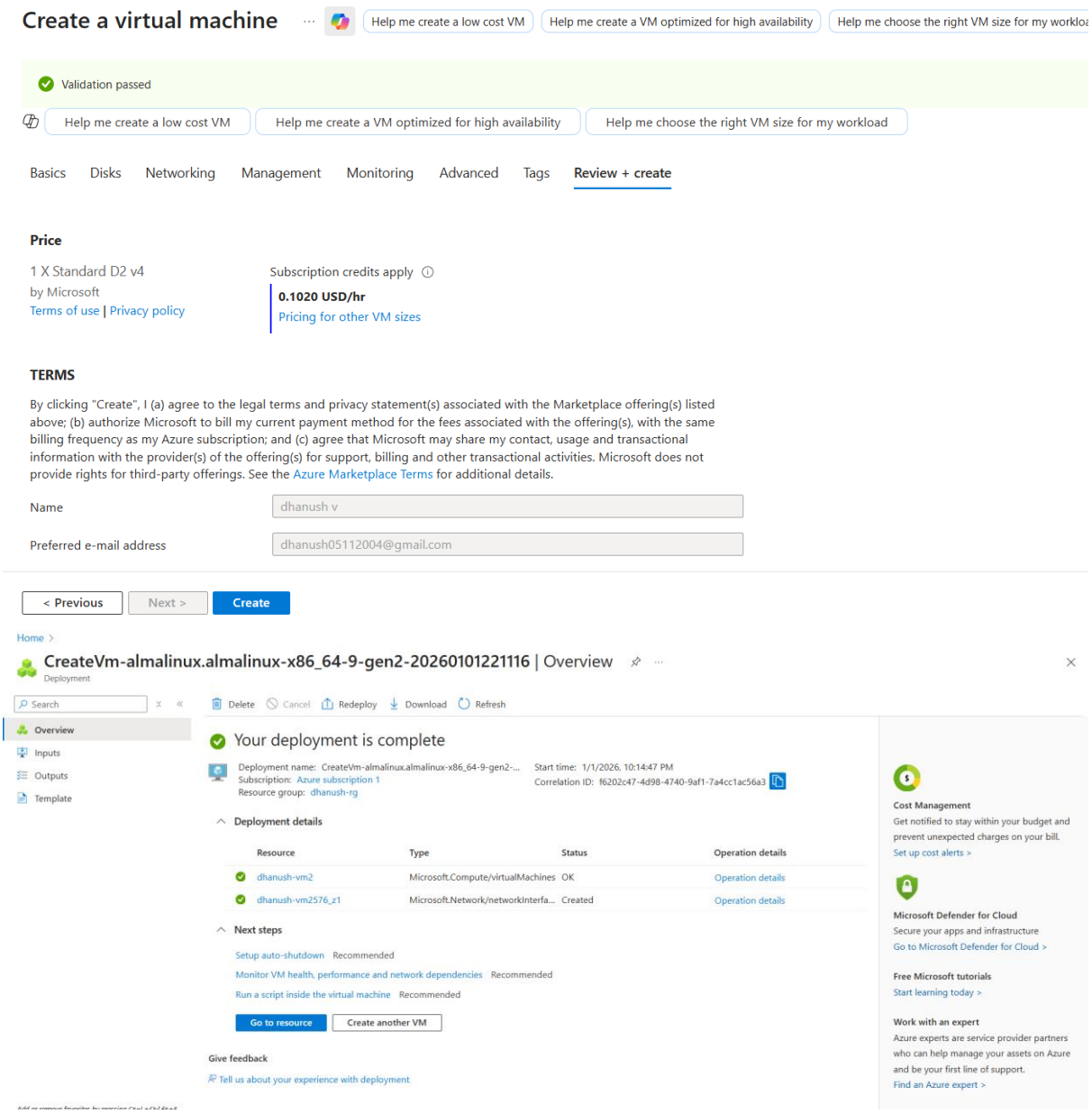
The selected subnet 'dhanush-private-subnet (10.0.2.0/24)' is already associated to a network security group 'dhanush-nsg'. We recommend managing connectivity to this virtual machine via the existing network security group instead of creating a new one here.

Configure network security group *
dhanush-nsg
(new) dhanush-vm2-nsg
dhanush-nsg
resource group: dhanush-rg

Delete NIC when VM is deleted ⓘ

Enable accelerated networking ⓘ

Load balancing

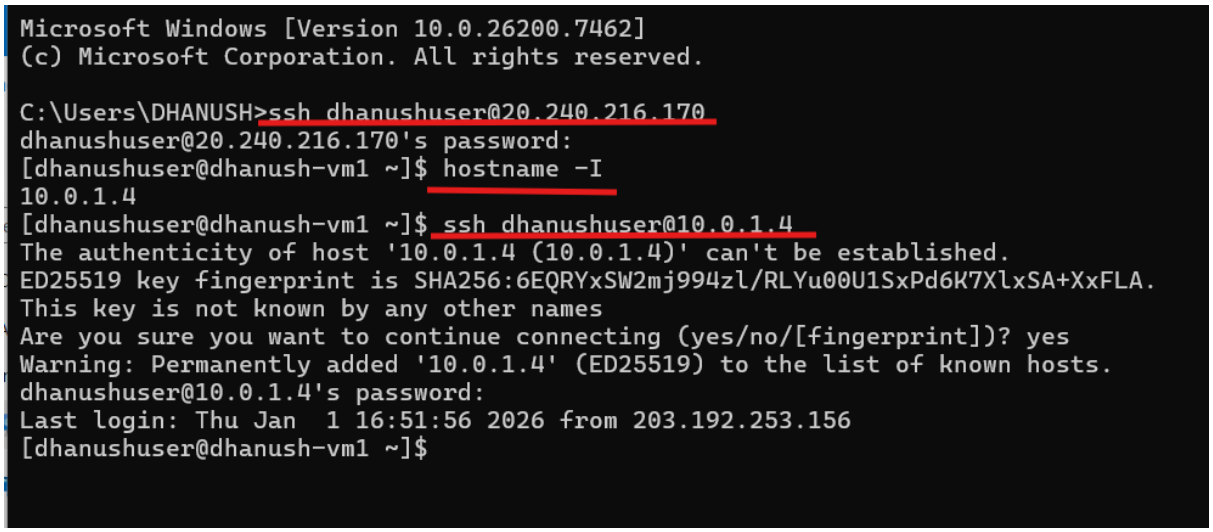


Step 6 : verification

Test Connectivity

1. From vm1 → ping vm2 → ✅ Works
2. From vm1 → ssh vm2 → ✅ Works

3. From vm1 → curl http://10.0.1.x → ❌ Blocked (due to Deny rule)
4. Remove deny-http rule → retry → ✅ Works



This way, every resource has a clear prefix:

- studentname-rg
- studentname-vnet
- studentname-subnet
- studentname-vm1, studentname-vm2
- studentname-vm1-pip
- studentname-nsg
- Rules: studentname-allow-ssh, studentname-allow-ping, studentname-deny-http